PROJECT PROPOSAL

Temporary Rental Space Management System of MFU

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BACHELOR OF ENGINEERING

IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

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A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE DEGREE OF
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Abstract

This senior project proposes the development of a web-based Temporary Rental Space Management System for Mae Fah Luang University. Designed to serve both internal university members and external organizations, the system addresses the inefficiencies of traditional, paper-based reservation methods. By fully digitizing the workflow, the platform aims to eliminate manual processing errors, reduce communication delays, and prevent scheduling conflicts.

The system provides a comprehensive online solution where users can check real-time availability, select customized add-ons, and process secure payments seamlessly. Simultaneously, administrators are equipped with a centralized dashboard for dynamic facility management, digital approval workflows, and data visualization. Ultimately, this system enhances the transparency, convenience, and overall operational efficiency of managing the university's commercial and non-commercial rental spaces.

Keyword: Space Management System, Web Application, Online Booking, Payment Integration, Mae Fah Luang University

CHAPTER 1
INTRODUCTION

1.1 Background and Rational

The Property Management Division of Mae Fah Luang University is primarily responsible for managing and monetizing a variety of temporary rental spaces. These facilities serve as a crucial source of revenue for the university, being utilized not only for internal academic activities but also for commercial rentals by external individuals, companies, and organizations.

However, the current space reservation process relies heavily on manual, paper-based procedures. This traditional approach is highly susceptible to human error, frequently resulting in scheduling conflicts, delayed communications, and significant complexities in managing financial transactions and official documentation with external clients.

To resolve these operational bottlenecks and enhance user convenience, this project proposes the development of a comprehensive web-based Temporary Rental Space Management System. By providing a fully digital platform that allows external users to easily book spaces and process payments online, the system will significantly reduce human errors and redundant manual tasks. Ultimately, this solution will empower the Property Management Division to maximize resource utilization and manage revenue generation with high accuracy and efficiency.

1.2 Objective

1.2.1 To develop a web-based temporary rental space management system for Mae Fah Luang University.
1.2.2 To enable both internal and external users to view room availability, select service add-ons, and process secure online payments.
1.2.3 To provide administrators with a centralized dashboard to efficiently manage facilities, approval workflows, and revenue tracking.

1.3 Scope

1.3.1 System Users

Users are categorized into three distinct authorization levels based on their relationship with Mae Fah Luang University (MFU):

- 1.3.1.1 Internal Users: Individuals operating within the MFU organization.
- 1.3.1.2 External Users: Individuals or entities outside the MFU organization.
- 1.3.1.3 Collaborative Users: Users involved in cross-organizational partnerships or activities.
- 1.3.1.4 System Administrator: Personnel responsible for overseeing facility data, managing reservation requests, and handling administrative workflows.

1.3.2 User Functional Scope:

- Availability Search & Advanced Booking: Check real-time room availability and submit advance reservation requests within specified timeframes.
- Service Customization (Add-ons): Select supplementary services or additional equipment (e.g., specific quantities of tables and chairs) during the booking process, with automated calculation of additional costs.
- Secure Payment Integration: Process transactions securely and seamlessly via an integrated payment gateway.
- Promotions and Incentives: Apply promotional discounts or special offers provided by the system during the reservation process.
- Reservation Management & History: Access personal booking logs, specify reservation purposes, track current statuses, and cancel existing bookings according to system policies.
- Status and Event Notifications: Receive automated alerts regarding booking confirmations, workflow status updates, and announcements for new promotional campaigns.
- Post-Service Feedback: Submit reviews and satisfaction ratings after completing the reservation lifecycle to provide service evaluations.
- **Secure Authentication:** Users can securely log in via Google OAuth. The system automatically assigns user roles (e.g., internal or external) based on their registered email domains, such as @mfu.ac.th or @lamduan.mfu.ac.th.
- **Secure Payment Integration:** Users can process secure and instant online payments via a third-party gateway, specifically utilizing dynamic PromptPay QR Codes.

1.3.3 Administrator Functional Scope:

- Facility & Rate Management: Maintain room records (Add/Edit/Disable) and manage pricing structures. This includes the ability to bulk import room details and room rates directly from spreadsheet datasets (Excel files).
- Administrative Dashboard & Data Visualization: Access a centralized analytics dashboard featuring comprehensive data visualization through interactive graphs and charts. This tool allows for monitoring real-time booking statistics, revenue tracking, and facility utilization, with the ability to drill down into specific data entries for detailed analysis.
- Approval Workflow and Documentation: Process reservation requests by updating statuses (Approve/Disapprove). This includes the ability to import executive authorization files and export official approval documents to serve as verified digital evidence.
- Promotion Management: Create, configure, and manage promotional campaigns to encourage system utilization.
- System Notifications & Broadcasts: Manage automated system alerts and broadcast new promotional information or critical updates to designated user groups.

1.4 Methodology

This project follows an iterative Software Development Life Cycle (SDLC) to ensure the system meets the requirements of Mae Fah Luang University.

- 1.4.1 Requirement Gathering and Analysis: Collect and analyze user requirements by reviewing existing manual reservation processes and examining documents from the Intellectual Property and Innovation Management Division.
- 1.4.2 System Design: Design the system architecture, user interface (UI), and database. This includes designing the relational database schema and mapping data relationships using PostgreSQL and pgAdmin4.
    - 1.4.2.1 Third-Party Service Integrations
      To enhance security and operational efficiency, the system incorporates the following services:
        - **Google OAuth Authentication:** The system uses Google OAuth for Single Sign-On (SSO), leveraging the university's existing Google Workspace. This eliminates the need for local password storage, thereby enhancing data security and user convenience.
        - **Payment Gateway (Opn Payments):** Opn Payments (formerly Omise) and 2C2P were evaluated for transaction processing. While 2C2P provides extensive offline payment channels, **Opn Payments** was selected as the optimal solution. It offers highly developer-friendly APIs, seamless integration with the project's tech stack (Node.js and Vue.js), and a competitive 1.65% transaction fee for PromptPay QR Codes. Additionally, its accessible sandbox environment allows for immediate testing without requiring complex corporate documentation, making it highly practical for academic development.
- 1.4.3 System Development: Implement the web application using a modern tech stack. The frontend will be developed using Vue.js and Tailwind CSS for a responsive user interface, while the backend API and server logic will be handled by Node.js.
- 1.4.4 System Testing and Evaluation: Conduct rigorous testing to identify and resolve bugs. This includes testing role-based access for internal and external users, validating the approval workflows, and ensuring the system prevents scheduling conflicts.
- 1.4.5 Documentation and Presentation: Prepare the formal Senior Project 1 and 2 documentation and defend the project before the examination committee.

1.5 Plan

- **Table 1.1** Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 examination										
System development										
System testing and evaluation										
Senior project 2 document preparation										
Senior Project 2 examination										

1.6 Expected Result

1.6.1 A fully functional and secure web-based room reservation system for Mae Fah Luang University.
1.6.2 A streamlined booking experience with integrated online payment, enhancing convenience for external commercial users.
1.6.3 Significantly reduced human errors and redundant paperwork in the reservation and financial tracking processes.
1.6.4 Improved operational efficiency and enhanced revenue generation capabilities for the Property Management Division.

1.7 Place and Equipment

1.7.1 Equipment

1.7.1.1 Hardware

- Personal computer or laptop

- Internet connection

1.7.1.2 Software

- Frontend: Vue.js, Tailwind CSS

- Backend: Node.js

- Database: PostgreSQL, pgAdmin4

1.7.1.3 Data set

- University room details and specifications

CHAPTER 2
LITERATURE REVIEW

This chapter reviews the existing literature, systems, and technologies relevant to the development of the Temporary Rental Space Management System. It evaluates the current booking methods at Mae Fah Luang University (MFU) and explores the core technologies and third-party integrations utilized in this project.

2.1 Review of the Existing MFU Room Booking System

Currently, the room booking process at Mae Fah Luang University relies significantly on manual workflows and basic internal platforms. These existing systems are primarily designed for internal academic use, lacking the flexibility to accommodate external commercial clients. Furthermore, the absence of an integrated online payment gateway forces users to complete transactions manually, leading to administrative delays.

To highlight the improvements proposed by this project, a comparison between the existing MFU system and the proposed system is presented in Table 2.1.

Table 2.1 Comparison between the Existing MFU Booking System

Features	Existing MFU Booking System	Proposed System
Primary Target Users	Internal users (Staff and Students)	Internal and External users
Workflow Format	Paper-based and manual approval	Fully automated and web-based
Authentication	Local university database login	Google OAuth (SSO)
Payment Method	Manual/Cash at the counter	Integrated Payment Gateway (PromptPay QR)
Service Add-ons	Manual request required	Automated add-on selection during booking
Real-time Status	Limited tracking capabilities	Real-time dashboard for users and admins
Revenue Generation	Not optimized for commercial rentals	Fully optimized for commercial leasing

2.2 Related Technologies and Integrations

To achieve the proposed features, several modern technologies and third-party services were reviewed and implemented:

- **2.2.1 Web Application Frameworks:** Vue.js and Tailwind CSS were selected for the frontend to provide a responsive and dynamic user interface. Node.js and PostgreSQL were chosen for the backend to ensure robust data management and high-performance server processing.
- **2.2.2 Single Sign-On (SSO) and Authentication:** Google OAuth is utilized to handle user authentication securely. This eliminates the need for localized password storage, leveraging Google's robust security infrastructure while providing a seamless login experience for users with existing university or personal Google accounts.
- **2.2.3 Payment Gateway Providers:** For the financial transaction module, Opn Payments (formerly Omise) was selected over other regional providers like 2C2P. Opn Payments offers a highly developer-friendly API, seamless integration with Node.js, and a highly competitive transaction fee for dynamic PromptPay QR Codes (1.65%). This makes it the most cost-effective and efficient solution for the university's Property Management Division.

CHAPTER 3

ANALYSIS AND DESIGN

ER-Diagram, Use-Case Diagram, User Interface(UI),
Swimlane Diagram

CHAPTER 4

CONCLUSION

4.1 Project Conclusion

The development of the Temporary Rental Space Management System successfully addresses the administrative bottlenecks faced by the Property Management Division at Mae Fah Luang University. By transitioning from a manual, paper-based workflow to a fully automated web application, the project significantly reduces human errors, prevents scheduling conflicts, and streamlines the overall facility management process.

The system empowers both internal and external users to easily browse room availability, select service add-ons, and process secure payments online via the Opn Payments gateway. Simultaneously, it provides administrators with a comprehensive dashboard to track revenue, manage approvals, and analyze facility utilization. Ultimately, this system fulfills its primary objective of optimizing resource management and enhancing the university's commercial revenue generation capabilities.

4.2 Limitations

Despite the successful implementation, the system has certain limitations. Currently, the system relies entirely on web-based access and does not have a dedicated native mobile application, which could further enhance user convenience. Additionally, the system currently handles payment verifications through standard API responses and does not integrate directly with physical IoT devices (such as smart door locks) to automatically grant physical access upon payment confirmation.

4.3 Future Recommendations

For future development, the following enhancements are recommended:

1. **Mobile Application Development:** Creating a cross-platform mobile application (iOS and Android) to provide users with push notifications and easier access to their booking QR codes.
2. **IoT Integration for Smart Access:** Integrating the booking system with IoT-enabled smart door locks. This would allow the system to automatically generate temporary digital access passcodes or NFC keys for users whose bookings and payments have been fully approved.
3. **Advanced Data Analytics:** Implementing machine learning algorithms within the admin dashboard to predict peak rental periods and suggest dynamic pricing models to maximize university revenue.